

SF2940: DISTRIBUTIONS AND THEIR CHARACTERISTIC FUNCTIONS

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a), a \in \mathbb{R}$	$p(a) = 1$	e^{ita}
Bernoulli	$\text{Be}(p); 0 \leq p \leq 1$	$p(0) = q, p(1) = p; q = 1 - p$	$q + pe^{it}$
Binomial	$\text{Bin}(n, p), n = 1, 2, \dots; 0 \leq p \leq 1$	$p(k) = \binom{n}{k} p^k q^{n-k}, k = 0, 1, 2, \dots, n; q = 1 - p$	$(q + pe^{it})^n$
Geometric	$\text{Ge}(p), 0 \leq p \leq 1$	$p(k) = pq^k, k = 0, 1, 2, \dots; q = 1 - p$	$\frac{p}{1 - qe^{it}}$
Poisson	$\text{Po}(\lambda), \lambda > 0$	$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$	$e^{\lambda(e^{it} - 1)}$

TABLE 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$U(a, b), a < b$	$f(x) = \frac{1}{b-a}, a < x < b$	$\frac{e^{itb} - e^{ita}}{it(b-a)}$
Exponential	$\text{Exp}(\lambda), \lambda > 0$	$f(x) = \frac{1}{\lambda} e^{-x/\lambda}, x > 0$	$\frac{1}{1 - \lambda it}$
Laplace	$L(\lambda), \lambda > 0$	$f(x) = \frac{1}{2\lambda} e^{- x /\lambda}, x \in \mathbb{R}$	$\frac{1}{1 + \lambda^2 t^2}$
Beta	$\beta(r, s), r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1} (1-x)^{s-1}, 0 < x < 1$	*
Normal	$N(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}, x \in \mathbb{R}$	$e^{i\mu t - \frac{1}{2}\sigma^2 t^2}$
Standard normal	$N(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, x \in \mathbb{R}$	$e^{-\frac{1}{2}t^2}$
Log-normal	$\text{LN}(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\log x - \mu)^2/\sigma^2}, x > 0$	*
Cauchy	$C(m, \eta), \eta > 0, m \in \mathbb{R}$	$f(x) = \frac{1}{\pi} \frac{\eta}{(x-m)^2 + \eta^2}, x \in \mathbb{R}$	$e^{imt - \eta t }$

TABLE 2. Continuous distributions

* indicates either a too complicated expression or no explicit expression exists.

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